

Be Productive!

CSX-4109 (Android Application Development)

https://github.com/AungThuraHein-Nov03/Pomodoro_Timer.git



Presented By – Aung Thura Hein (6726135)

What is Pomodoro Technique?

- **The 25 minutes sprint**

You pick one task and focus on it exclusively for 25 minutes.

- **The Five minutes refresh**

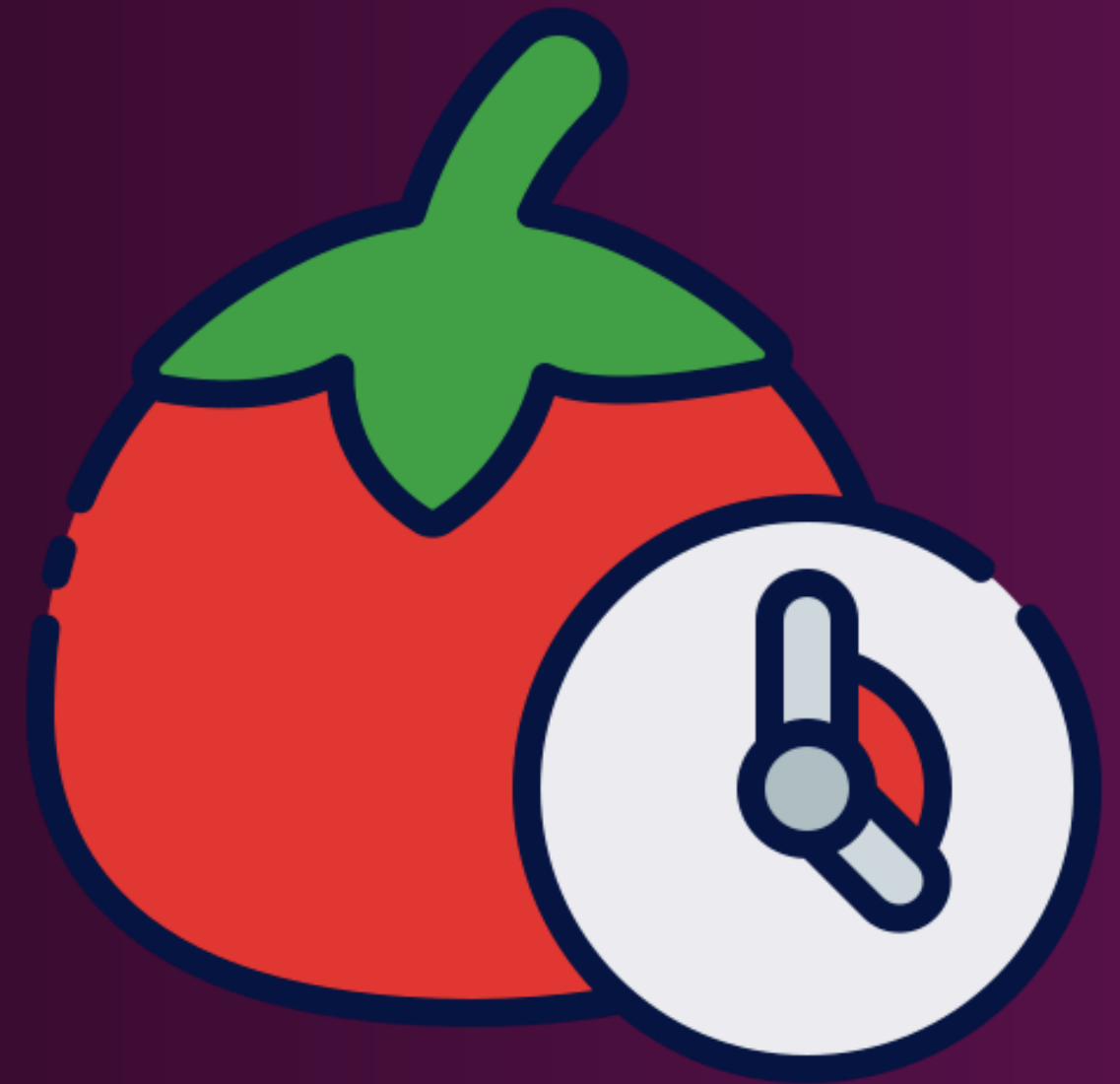
When the timer dings, there is a mandatory 5-minute break.

- **The "Tomato" Origin**

It was named by Francesco Cirillo in the late 1980s after the tomato-shaped kitchen timer he used to track his work as a kid.

- **The Long Reward**

After completing four Pomodoros , the break extends to 15 minutes.



The Problem we're addressing

Challenges

- Procrastination
- Difficulty starting
- Lack of motivation
- No tracking

Solution

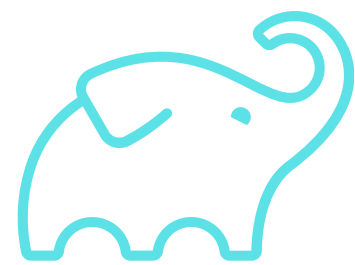
- Pomodoro Technique
- Improves Focus
- Reduces Fatigue
- Proven since 1980s

Architecture and Tech Stack



Kotlin

- Null safety
- Concise Syntax
- Google Supports
- Coroutine



Gradle

- Type safe
- Kotlin DSL
- Version Catal



UI

- ViewBinding
- Material3
- Constraint Layout
- BarChartView

Architecture and Tech Stack



Database

- SQLiteOpenHelper
- SharedPreferences
- Data classes
- Complex SQL



Platform

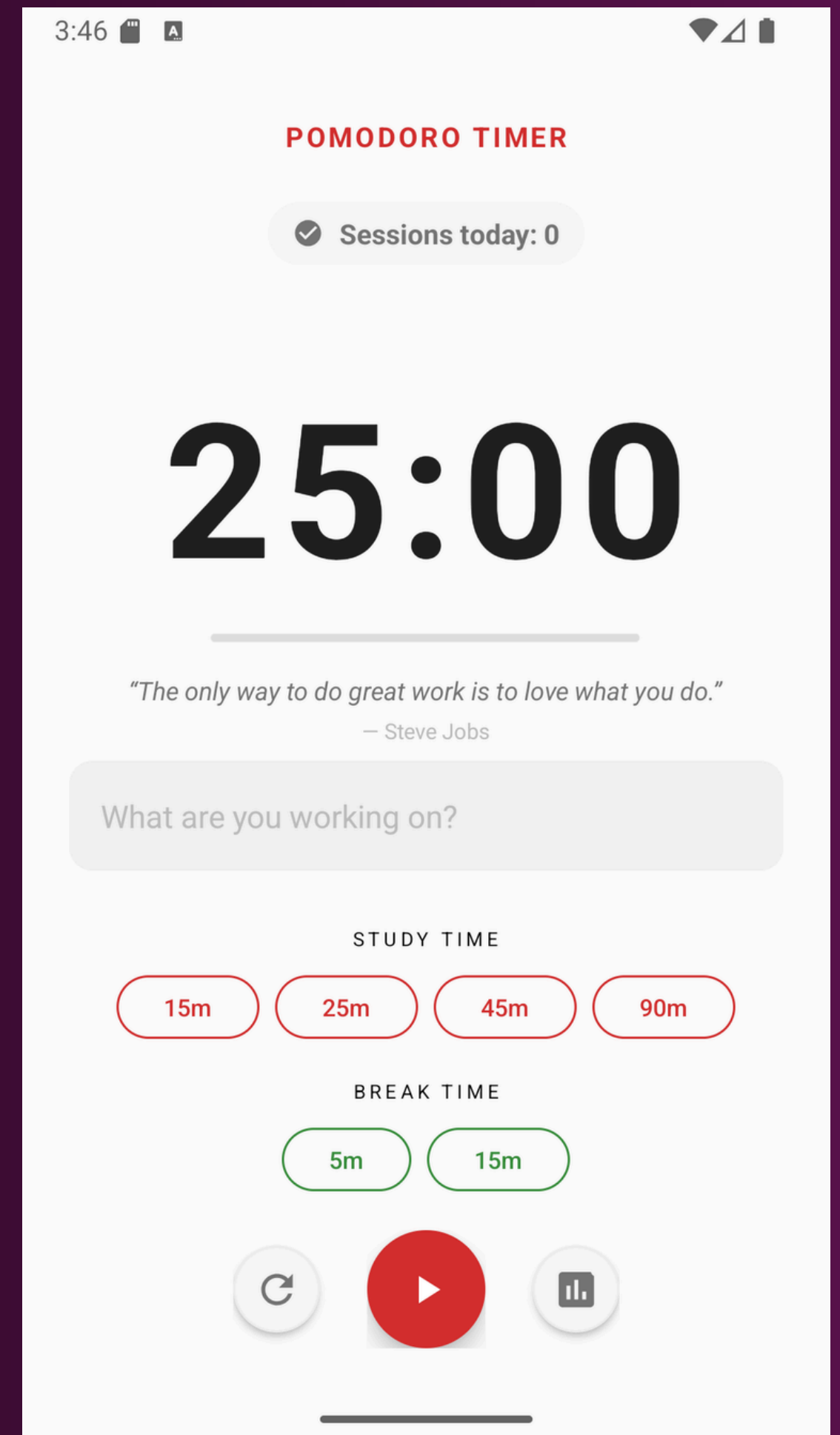
- Min SDK: 26
- Target SDK: 36
- Modern Compat.



Home Page

MainActivity.kt + activity_main.xml

- ✓ CountdownTimer for precision
- ✓ State management (running/paused/ready)
- ✓ Presets disabled during active sessions
- ✓ AlertDialog before reset
- ✓ Auto-launch BreakActivity
- ✓ SharedPreferences persistence





Database

SessionDatabaseHelper.kt

- insertSession (timestamp, duration, task)
- getTotalWorkMinutes()
- getBestStreakDays()
- getDailyMinutes(days)
- getRecentSessions(limit)

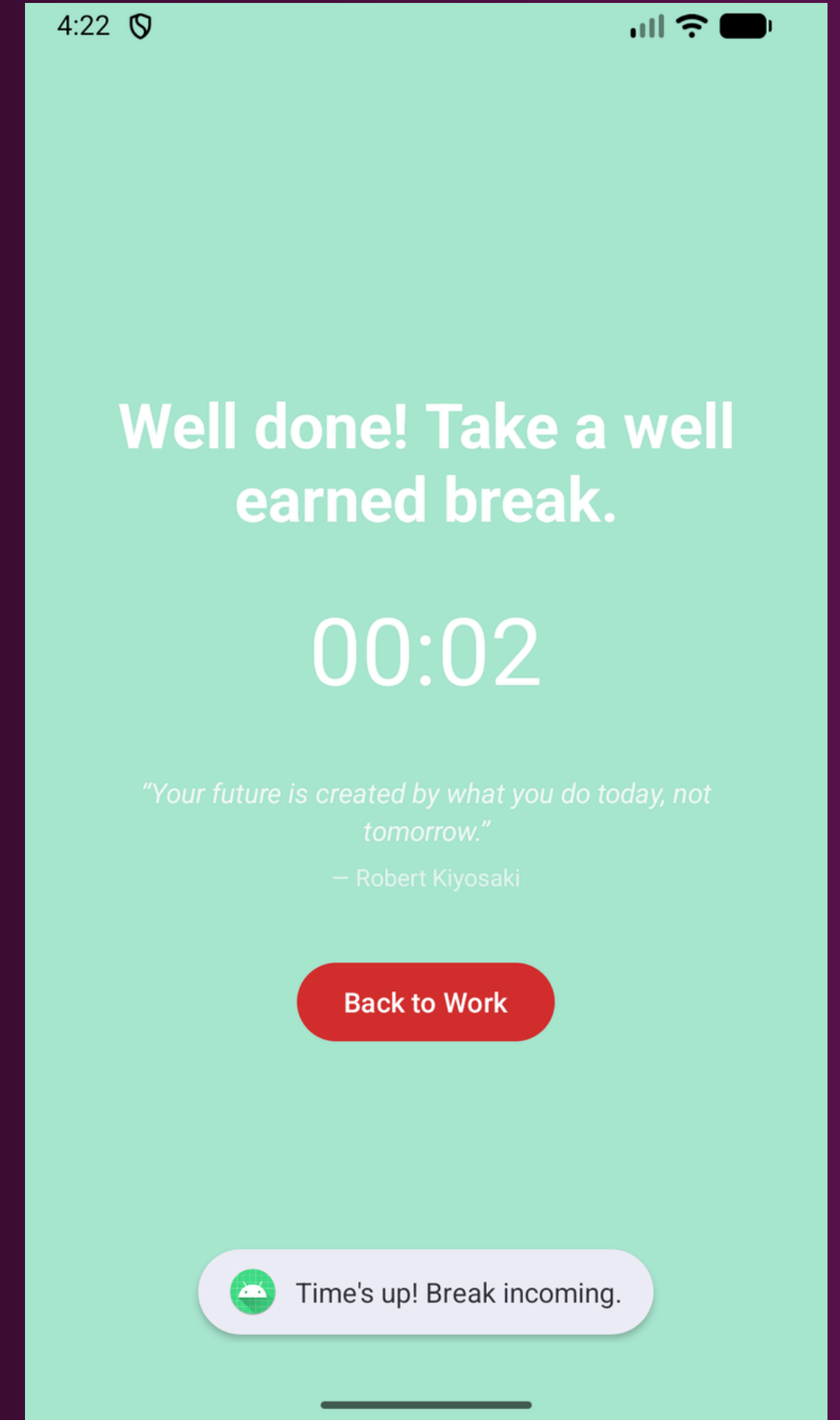
```
1 Usage
override fun onCreate(db: SQLiteDatabase) {
    db.execSQL(
        sql = """CREATE TABLE $TABLE_SESSIONS (
            $COL_ID INTEGER PRIMARY KEY AUTOINCREMENT,
            $COL_DATE TEXT NOT NULL,
            $COL_TASK_NAME TEXT NOT NULL,
            $COL_DURATION INTEGER NOT NULL,
            $COL_TIMESTAMP INTEGER NOT NULL
        )"""
    )
}
```



Break Page

BreakActivity.kt + activity_break.xml

- ✓ Automatic launching via Intent
- ✓ No manual switching required
- ✓ Clean work/break separation
- ✓ Easy return (single tap)
- ✓ No timer state corruption
- ✓ Motivational quote displayed

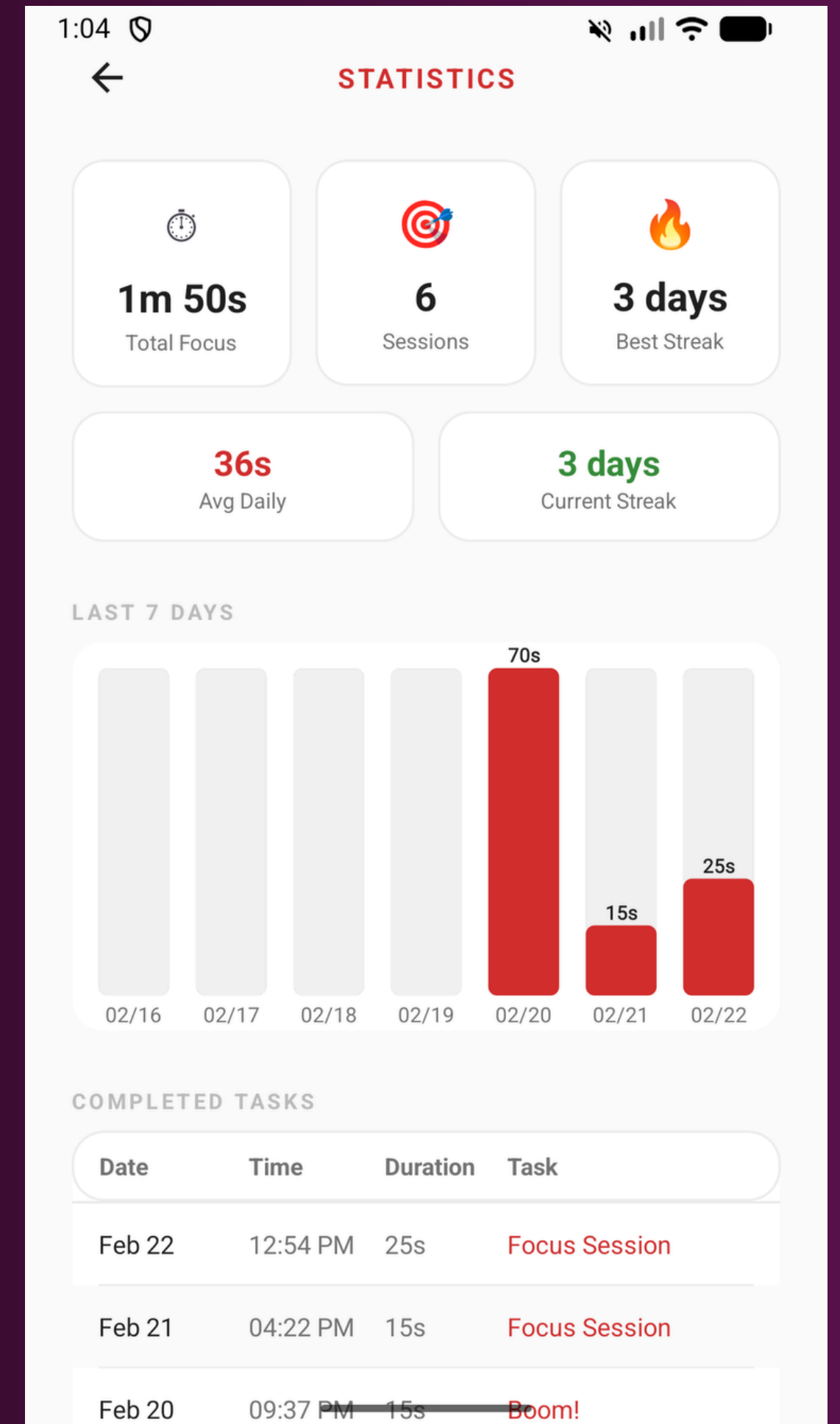




Stats Page

StatisticsActivity.kt + activity_statistics.xml

- ✓ Summary Card
- ✓ Custom BarChart
- ✓ Task Table





User Experience

- ✓ 25 productivity-focused quotes
- ✓ A no-repeat algorithm
- ✓ Notification
- ✓ Visual feedback

2 Usages

```
private var lastIndex = -1
```

2 Usages

```
fun getRandom(): Quote {  
    if (quotes.size <= 1) return quotes.first()  
    var index: Int  
    do {  
        index = quotes.indices.random()  
    } while (index == lastIndex)  
    lastIndex = index  
    return quotes[index]  
}  
}
```



Challenges and Solutions

- Calculating daily statistics from hundreds of session
- Building BarChartView
- State Management
- Optimized SQL queries (GROUP BY)
- Limited recent sessions and displays
- Studied Canvas API documentation
- Track work, rest, break modes



Future Updates

Widget & Darkmode

- Widgets
- Proper dark mode
- Color palettes

Advanced Analytics

- Calendar view
- Productivity
- Gamification
- Monthly/yearly

February 25th

CSX-4109 (Android App Development)

A SINCERE
THANK YOU TO
All Listeners



[HTTPS://GITHUB.COM/AUNGTHURAHEIN-NOV03/POMODORO_TIMER.GIT](https://github.com/AUNGTHURAHEIN-NOV03/POMODORO_TIMER.GIT)